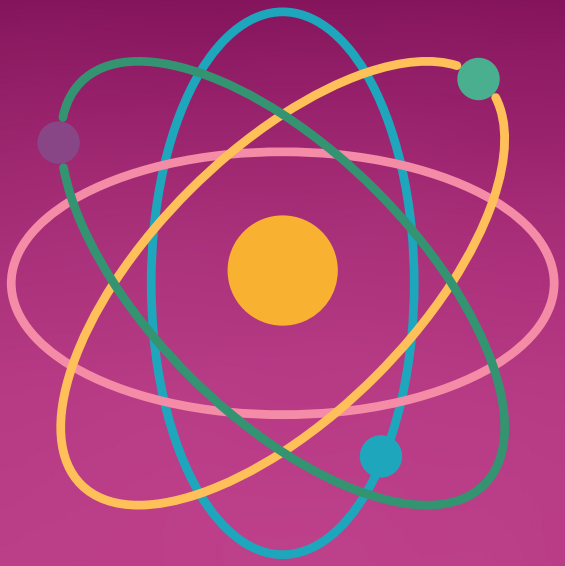
ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

path to success



PHYSICAL CHEMISTRY

JEE (MAIN+ADVANCED)

ENTHUSIAST COURSE

EXERCISE

Thermochemistry

English Medium

EXERCISE # S-I

Miscellaneous

Q.1 Lime is made commercially by decomposition of limestone, $CaCO_3$. What is the change in internal energy (in kJ) when 1.00 mole of solid $CaCO_3$(V = 34 ml) absorbs 180 kJ of heat and decomposes at certain temperature against a pressure of 1.0 bar to give solid CaO.(Volume = 16 ml) and CO_2(g)(V = 20 L). [Give your answer as nearest integer] **TC0023**

Q.2 For the gaseous reaction:

$3A + 2B \longrightarrow 4C$; $\Delta H = -300$ Cal/mol

If 5 moles of A are mixed with 4 moles of B at 300K, the magnitude of work (in cal) involved in reaction is. **TC0025**

Q.3. When the following reaction was carried out in a bomb calorimeter, ΔU is found to be –742.7 kJ/mol of NH_2CN(s) at 300 K.

$$NH_2CN(s) + \frac{3}{2}O_2\ (g) \longrightarrow N_2(g) + CO_2(g) + H_2O(l)$$

Calculate ΔH_{300K} (in kJ/mole) for the reaction. (R = 8 J/mole-K). **TC0093**

Q.4. For the reaction : $2A\ (g) + 3B\ (g) \longrightarrow 4C\ (g) + D\ (\ell)$; $\Delta H = -20$ KJ/mole
Find the heat gained (in kJ) when 0.4 mole of A reacts with excess of B in a closed rigid container. The temperature is constant at 300 K. [R=8.3 J/K mole]

TC0094

Q.5 Ethyl chloride (C_2H_5Cl), is prepared by reaction of ethylene with hydrogen chloride :

$$C_2H_4(g) + HCl(g) \longrightarrow C_2H_5Cl(g) \qquad \Delta H = -72.3 \text{ kJ/mol}$$

What is the value of ΔU (in kJ), if 70 g of ethylene and 73 g of HCl are allowed to react at 300 K ?

TC0004

Q.6 Determine the amount of heat (in kcal) lost to surroundings at constant volume when 0.5 mol of N_2 & 1.5 mol of H_2 reacted according to equation at 300 K.
$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$; $\Delta_rH_{300} = -380$ kcal/mol
(Given : R = 2 cal / mol-K) **TC0003**

Q.7 The enthalpy change for the reaction of 50 ml of ethylene with 50.0 ml of H_2 at 1.5 bar pressure is $\Delta H = -0.31$ KJ. Find ΔU (in kJ) for this process ? **TC0001**

Enthalpy of formation and combustion

Q.8 Find standard enthalpy of formation of ammonia gas in kJ/mol -

Given : $N_2H_4(g) + H_2(g) \longrightarrow 2NH_3(g)$; $\Delta_rH^0 = -40 kJ/mol$

$\Delta H_f^\circ[N_2H_4(g)] = -120 kJ/mol$

TC0095

Q.9 The enthalpy change for the reaction $C_3H_8(g) + H_2(g) \longrightarrow C_2H_6(g) + CH_4(g)$ at 25° C is – 55.7 kJ/mol. Calculate the enthalpy of combustion of C_2H_6(g) (in kJ/mole). The enthalpy of combustion of H_2, C_3H_8 & CH_4 are –285.7, – 2220 & –890.0 kJ/mol respectively.

TC0096

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Q.10 Calculate standard enthalpy of formation of carbon–di–sulphide (*l*) (in kJ/mole). Given the standard enthalpy of combustion of carbon (s) , sulphur (s) & carbon–di–sulphide (*l*) are : – 393, – 293 and – 1108 kJ mol^{-1} respectively.

TC0097

Q.11 At 300 K, the standard enthalpies of formation of $C_6H_5COOH(s)$, CO_2 (g) & H_2O (*l*) are – 408, – 393 & –286 kJ mol^{-1} respectively . Calculate the heat of combustion (in kJ/mole) of benzoic acid at : (i) constant pressure (ii) constant volume. **TC0008**

Q.12 When 2 moles of $C_2H_6(g)$ are completely burnt, 3120 kJ of heat is liberated . Calculate the enthalpy of formation (in kJ/mol) of $C_2H_6(g)$. Given $\Delta_f H$ for $CO_2(g)$ & H_2O (*l*) are – 395 & – 286 kJ respectively.

TC0006

Q.13 From the following data at 25°C, Calculate the standard enthalpy of formation of FeO(s) and of $Fe_2O_3(s)$.

Reaction	$\Delta_r H^o$ **(kJ/mole)**
(1) $Fe_2O_3(s) + 3C(graphite) \rightarrow 2Fe(s) + 3CO(g)$	492
(2) $FeO(s) + C(graphite) \rightarrow Fe(s) + CO(g)$	155
(3) $C(graphite) + O_2(g) \rightarrow CO_2(g)$	–393
(4) $CO(g) + 1/2\ O_2(g) \rightarrow CO_2(g)$	–282

TC0007

Relationship between ΔH & ΔU

Q.14 Determine ΔH for the following reaction at 500K and constant pressure :

$CO(g) + H_2O(g) \rightarrow CO_2(g) + H_2(g)$

use the following data :

Substance	C_p**(J/mol K)**	$\Delta_f H$**(300K) (kJ/mol)**
CO (g)	29	–110
H_2O (g)	33	–241
CO_2 (g)	37	–393
H_2 (g)	29	0

TC0005

Q.15. 0.16 g of methane was subjected to combustion at 27° C in a bomb Calorimeter . The temperature of Calorimeter system (including water) was found to rise by 0.5° C . Calculate the heat of combustion of methane at [Heat capacity of carlorimeter & its contents = 17.7 kJ/mole]
(i) constant volume (ii) constant pressure. [Give your answer as nearest integer]

TC0098

Q.16. One gm sample of NH_4NO_3 (s) is decomposed in a bomb calorimeter (constant volume). The temperature of the calorimeter system falls by 6K. If the heat capacity of system is 1.25 kJ/K, what is the molar enthalpy of decomposition of $NH_4NO_3(s)$ (in kJ/mole) at 300 K.
[R = 8.3 JK^{-1} mol^{-1}]. $NH_4NO_3(s) \rightarrow N_2O(g) + 2H_2O(l)$.

TC0099

Q.17 0.821 *l* sample of a mixture of $CH_4(g)$ & $O_2(g)$ measured at 27° C & 760 torr was allowed to react at constant pressure in a calorimeter which together with its contents had a heat capacity of 1200 Cal/K. The complete combustion of methane to CO_2 & H_2O caused a temperature rise, in the calorimeter, of 0.25 K. What was the mole percent of CH_4 in the original mixture?
$\Delta H^o_{comb}(CH_4) = -200$ kcal mol^{-1} . **TC0010**
The thermal capacity of Calorimeter system is 17.7 kJ K^{-1} . (R = 8.313 mol^{-1} K^{-1})

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Bond Enthalpy

Q.18 For the reaction : $N_2H_4(g) \rightarrow N_2H_2(g) + H_2(g)$; $\Delta_r H^\circ = 109$ kJ /mol
Calculate the bond enthalpy of N = N.
Given : B.E. (N – N) = 163 kJ/mol, B.E. (N–H) = 391 kJ/mol, B.E. (H–H) = 436 kJ/mol

TC0100

Q.19 Using bond enthalpy data, calculate enthalpy of formation of gaseous isoprene .

$$5\,C(s) + 4\,H_2(g) \longrightarrow H_2C = \underset{\displaystyle CH_3}{\overset{\displaystyle |}{C}} - \underset{\displaystyle H}{\overset{\displaystyle |}{C}} = CH_2(g)$$

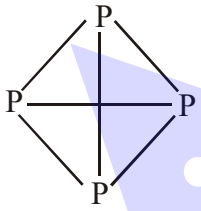
Given : C–H = 98.8 kcal/mol ; H – H = 104 kcal/mol ; C–C = 83 kcal/mol ; C = C = 147 kcal/mol & $C(s) \rightarrow C(g) = 171$ kcal/mol.

TC0101

Q.20 Compute the enthalpy of formation of liquid methyl alcohol in kJ mol^{-1}, using the following data. Enthalpy of vaporisation of liquid $CH_3OH = 38$ kJ/ mol .
Enthalpy of formation of gaseous atoms from the elements in their standard states are
$H \rightarrow 218$ kJ / mol ; $C \rightarrow 715$ kJ / mol ; $O \rightarrow 249$ kJ / mol.
Bond Enthalpies
C – H → 415 kJ / mol ; C – O → 356 kJ / mol ; O – H → 463 kJ / mol **TC0011**

Q.21 The polymerisation of ethylene to linear polyethylene is represented by the reaction
$n\,CH_2 = CH_2(g) \rightarrow (-CH_2 - CH_2)_n(g)$
where n has a large integral value . Given that the average enthalpies of bond dissociation for C=C & C–C at 298 K are + 590 & + 331 kJ mol^{-1} respectively. Calculate the enthalpy of polymerisation per mole of ethylene (in kJ) at 298 K . **TC0013**

Q.22 White phosphorus is a tetra-atomic solid $P_4(s)$ at room temperature.



Find average (P –P) bond enthalpy in kJ/mol.
Given : $\Delta H_{sublimation}$ of P_4 (s) = 59 kJ/mol
$\Delta H_{atomisation}$ of P_4 (s) = 1265 kJ/mol **TC0014**

Q.23 Calculate the magnitude of resonance enthalpy (in kJ/mole) of $CO_2(g)$ from following data

$\Delta H^\circ_{combustion}[C_{(graphite)}] = -390$ kJ/mol

$\Delta H^\circ_{atomisation}[C_{(graphite)}] = 715$ kJ/mol

$\Delta H^\circ_{B.E.}[O=O] = 500$ kJ/mol

$\Delta H^\circ_{B.E.}[C=O] = 800$ kJ/mol **TC0017**

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Q.24 Calculate enthalpy of combustion of propane [C_3H_8 (g)] in **kJ/mol** at 298 K.

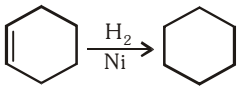
Given : B.E. (O = O) = 498 kJ/mole ; B.E. (C = O) = 804 kJ/mole

B.E. (C–H) = 410 kJ/mole ; B.E. (O–H) = 464 kJ/mole

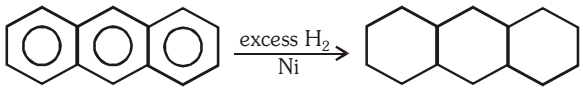
B.E. (C–C) = 345 kJ/mole ; Resonance energy of CO_2 (g) = – 143 kJ/mole

$\Delta H_{vaporization}$ (H_2O, *l*) = 41 kJ/mole **TC0015**

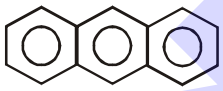
Q.25 Use the following data to answer the questions below :



; $\Delta H = -28.6$ Kcal mol^{-1} ;



; $\Delta H = -116.2$ Kcal mol^{-1}

Calculate the resonance energy of anthracene,  **TC0016**

Other types of Enthalpy of reaction

Q.26 Calculate the electron gain enthalpy of fluorine atom (in kJ/mole) using the following data . (All the values are in kJ mol^{-1} at 25° C). ΔH_{diss} (F_2) = 160, $\Delta_f H$ (NaF(s)) = – 571, I.E. [Na(g)] = 494, ΔH_{sub} [Na(s)] = 101 , Lattice enthalpy of NaF(s) = 894.

TC0102

Q.27 The enthalpy of solution of anhydrous $CuSO_4$ is –16 kcal/mol and that of $CuSO_4.5H_2O$ is 3 kcal/mol Calculate the enthalpy of hydration of $CuSO_4$ (in kcal/mol)

TC0103

Q.28 If the enthalpy of formation of HCl (g) and Cl^- (aq) are –90 kJ/mole and – 170 kJ/mol, find the enthalpy of solution of hydrogen chloride gas (in kJ/mol). **TC0021**

Q.29 Calculate the enthalpy of ionisation of weak acid H_2A($H_2A \rightarrow 2H^+ + A^{2-}$) in Kcal/ mol, if enthalpies of neutralisation of HCl and H_2A by a strong base are –14 Kcal/eq and –11 Kcal/eq respectively.

TC0104

Q.30 Enthalpy of neutralization of H_3PO_3 acid is -106.68 kJ/mol using NaOH. If enthalpy of neutralization of HCl by NaOH is –55.84 kJ/mol. Calculate $\Delta H_{ionization}$ of H_3PO_3 into its ions (in kJ/mol)

TC0105

Q.31 Two solutions initially at 25°C were mixed in an adiabatic constant pressure calorimeter. One contains 400 ml of 0.2 M weak monoprotic acid solution. The other contain 100 ml of 0.80 M NaOH. After mixing temperature increased to 26.2 °C. How much heat is evolved in the neutralization of 1 mole of acid (in kJ) ? Assume density of solution 1.0 g/cm^3, and specific heat of solution 4.2 J/g-K. Neglect heat capacity of the calorimeter. **TC0106**

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EXERCISE # S-II

Q.1 The enthalpy of formation of $C_2H_5OH(l)$ is – 66 k Cal/mol . The enthalpy of combustion of $CH_3–O–CH_3$ is – 348 k Cal/mol . Given that the enthalpies of formation of $CO_2(g)$ and $H_2O(l)$ are – 94 k Cal/mol & – 68 kcal/mol respectively, calculate ΔH (in kcal/mole) for the isomerisation of ethanol to methoxymethane. All data are at 25° C. **TC0026**

Q.2 Calculate the mass of mercury (in grams) which can be liberated from HgO at 27° C by the treatment of excess HgO with 9 kJ of heat at (a) constant pressure, (b) constant volume

Given : ΔH_f° (HgO, s) = – 90 kJ mol^{-1} & Molar mass of (Hg) = 200 g mol^{-1} . **TC0027**

Q.3 A stoichiometric mixture of ferric oxide & Al is used as solid rocket fuel . Calculate the fuel value ((in kcal/gram) & fuel value (in kcal/cc) of the mix . Enthalpy of formation & densities are :

ΔH_f° (Al_2O_3) = – 399 k Cal/mole ; ΔH_f° (Fe_2O_3) = – 199 kcal / mole,

density of Fe_2O_3 = 5.2 g/cc ; density of Al = 2.7 g/cc. **TC0028**

Q.4 Calculate the enthalpy change for the reaction (in kcal/mole) : $XeF_4 \longrightarrow Xe^+ + F^- + F_2 + F$.

The average Xe–F bond enthalpy is 34 kcal/mol, first I.E. of Xe is 279 kcal/mol, electron affinity of F is 85 kcal/mol & bond dissociation enthalpy of F_2 is 38 kcal/mol. **TC0029**

Q.5 Calculate the bond enthalpy of the O–H bond in water (in kJ/mole) at 298K using the data/information given below :-

(A) $\Delta_f H^0[H(g)] = 218$ kJ/mol

(B) $\Delta_f H^0[O(g)] = 249.2$ kJ/mol

(C) $\Delta_f H^0[H_2O(g)] = –241.8$ kJ/mol

The bond enthalpy of the O–H bond in water is defined as one-half of the enthalpy change for the reaction $H_2O(g) \rightarrow 2H(g) + O(g)$.

Also, determine the ΔU of the O–H bond in water at 298K (in kJ/mole) . Assume ideal gas behaviour.

Given : $(8.314 \times 0.298 = 2.5)$ **TC0091**

Q.6 (i) From the following data :

Enthalpy of formation of C_2H_6 = – 83 kJ/mol

Enthalpy of sublimation of graphite = 719 kJ/mol

Enthalpy of bond dissociation of H_2 = 435 kJ/mol

C–H bond enthalpy = 414 kJ/mol

Calculate the bond enthalpy of C – C (in kJ/mole)

(ii) By using data of above question calculate bond enthalpy of C≡N (in kJ/mole) if enthalpy of formation of CH_3CN = 87 kJ/mol and enthalpy of bond dissociation of nitrogen = 945 kJ/mol

TC0092

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Q.7 Calculate the enthalpy change (in kcal/mole) when infinitely dilute solution of $CaCl_2$ and Na_2CO_3 are mixed $\Delta_fH°$ for $Ca^{2+}(aq)$, $CO_3^{2-}(aq)$ and $CaCO_3(s)$ are – 129, – 161, – 288 kcal mol^{-1} respectively.

TC0030

Q.8 The enthalpy of formation of ethane(g), ethylene(g) and benzene(g) from the gaseous atoms are –2840, –2275 and –5530 $kJmol^{-1}$ respectively. Calculate the magnitude of resonance energy of benzene (in kJ/mole) . The bond enthalpy of C–H bond is given as equal to + 410 kJ/mol.

TC0031

Q.9 Calculate the enthalpy of combustion of methyl alcohol (*l*) (in kJ/mole) at 298 K from the following data

Bond	C – H	C – O	O – H	O = O	C = O
Bond Enthalpy($kJ\ mol^{-1}$)	414	351.5	464.5	494	711

Resonance energy of CO_2 = – 143 kJ mol^{-1}

Latent heat of vaporisation of methyl alcohol = 35.5 kJ mol^{-1}.

Latent heat of vaporisation of water = 40.6 kJ mol^{-1}. **TC0032**

Q.10 The bond enthalpies of C–C, C=C & C≡C bonds are 348, 610 & 835 kJ/mol respectively at 298K & 1 bar. What is of the enthalpy change of polymerisation at 298K & 1 bar per mole of 2-butyne (in kJ/mole) ?

$nCH_3–C \equiv C–CH_3(g) \rightarrow (–CH_2–CH=CH–CH_2–)_n\ (g)$

['n' is a large integral value] **TC0033**

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EXERCISE # O-I

Relationship of ΔH & ΔU

Q.1 For a reaction, $2X(s) + 2Y(s) \rightarrow 2C(\ell) + D(g)$

The q_p at 27°C is – 28 Kcal mol^{-1}, the q_V is ---------- Kcal mol^{-1}

(A) – 27.4 (B) + 27.4 (C) – 28.6 (D) 28.6 **TC0034**

Q.2. On the basis of the following thermochemical data :

$H_2O(\ell) \rightarrow H^+(aq) + OH^-(aq)$; $\Delta H = 57.32$ kJ

$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(\ell)$; $\Delta H = -286.20$ kJ

The value of enthalpy of formation of OH^- ion at 25°C is :- **[AIEEE-2009]**

(A) +228.88 kJ (B) –343.52 kJ (C) –22.88 kJ (D) –228.88 kJ

TC0107

Q.3 For which reaction will $\Delta H = \Delta U$?

(A) $H_2(g) + Br_2(g) \rightarrow 2\,HBr(g)$ (B) $C(s) + 2\,H_2O(g) \rightarrow 2H_2(g) + CO_2(g)$

(C) $PCl_5(g) \rightarrow PCl_3(g) + Cl_2(g)$ (D) $2CO(g) + O_2(g) \rightarrow 2\,CO_2(g)$

TC0108

Q.4 A mixture of 2 moles of carbon monoxide and one mole of oxygen in a closed vessel is ignited to get carbon dioxide. If ΔH is the enthalpy change and ΔU is the change in internal energy, then :-

(A) $\Delta H > \Delta U$ (B) $\Delta H < \Delta U$ (C) $\Delta H = \Delta U$ (D) Not definite

TC0109

Q.5 What amount of heat energy (kJ) is released in the combustion of 12.0 g of C_3H_4 at 1atm constant pressure.

$C_3H_4(g) + 4\,O_2(g) \rightarrow 3\,CO_2(g) + 2\,H_2O(l)$; $\Delta H^o = -1939$ kJ

(A) 696.3 (B) 1939 (C) 6463.3 (D) 581.7 **TC0040**

Q.6 $NH_3(g) + 3Cl_2(g) \rightarrow NCl_3(g) + 3HCl\,(g)$; ΔH_1

$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$; ΔH_2

$H_2(g) + Cl_2(g) \rightarrow 2HCl\,(g)$; ΔH_3

The enthalpy of formation of NCl_3 (g) in the terms of ΔH_1, ΔH_2 and ΔH_3 is

(A) $\Delta H_f = -\Delta H_1 + \frac{\Delta H_2}{2} - \frac{3}{2}\Delta H_3$ (B) $\Delta H_f = \Delta H_1 + \frac{\Delta H_2}{2} - \frac{3}{2}\Delta H_3$

(C) $\Delta H_f = \Delta H_1 - \frac{\Delta H_2}{2} - \frac{3}{2}\Delta H_3$ (D) $\Delta H_f = \Delta H_1 + \Delta H_2 - \Delta H_3$ **TC0041**

Q.7 Consider the reaction at 300 K

$H_2(g) + Cl_2(g) \longrightarrow 2HCl(g)$ $\Delta H = -185$kJ/mol

Calculate ΔU if 3 mole of H_2 completely react with 3 mole of Cl_2 to form HCl.

(A) 0 (B) –185 kJ (C) 555 kJ (D) – 555 kJ

TC0035

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Q.8 Ethanol can undergoes decomposition to form two sets of products

$$C_2H_5OH\,(g) \longrightarrow \begin{cases} \xrightarrow{1} C_2H_4(g) + H_2O\,(g) \quad ; \quad \Delta H^\circ = 45 \text{ kJ} \\ \xrightarrow{2} CH_3CHO(g) + H_2\,(g) \; ; \quad \Delta H^\circ = 63 \text{ kJ} \end{cases}$$

if the molar ratio of C_2H_4 to CH_3CHO is 8 : 1 in a set of product gases, then the enthalpy involved in the decomposition of 1 mole of ethanol is

(A) 423 kJ (B) 47 kJ (C) 61 kJ (D) 549 kJ **TC0050**

Q.9 Given the correct order of initials **T** (true) or **F** (false) for following statements.

(i) For the reaction $CaCO_3$ (Calcite) $\rightarrow$ $CaCO_3$ (aragonite) ; $\Delta H = -1127.75$ kJ/mol, then Calcite form is more stable at standard conditions.

(ii) For the reaction,

(a) $C(\text{diamond}) + 2H_2\,(g) \rightarrow CH_4(g)$; ΔH_1

(b) $C(g) + 4H(g) \rightarrow CH_4(g)$; ΔH_2

then more heat is evolved in reaction (b)

(iii) $\Delta_f H^\circ\,(I_2, g) = \Delta_{sub} H\,[I_2, s]$ at 25°C

(iv) For the exothermic reaction $2Ag(s) + 1/2\,O_2(g) \rightarrow 2Ag_2O(s)$ at 298 K. $\Delta H < \Delta U$

(A) FTTT (B) TTFT (C) TFTF (D) TTTT **TC0047**

Q.10 Use the given standard enthalpies of formation to determine the enthalpy of reaction of the following reaction : $TiCl_4(g) + 2\,H_2O(g) \rightarrow TiO_2(g) + 4\,HCl(g)$

$\Delta H^o_f\,TiCl_4(g) = -763.2$ kJ/mole $\Delta H^o_f\,TiO_2(g) = -944.7$ kJ/mole

$\Delta H^o_f\,H_2O(g) = -241.8$ kJ/mole $\Delta H^o_f\,HCl(g) = -92.3$ kJ/mole

(A) – 278.1 kJ (B) + 369.2 kJ (C) + 67.1 kJ (D) – 67.1kJ

TC0110

Q.11. For which species $\Delta H_{formation} \neq 0$

(A) $H^+_{(aq.)}$ (B) $Br_{2(g)}$ (C) $I_{2(s)}$ (D) $C_{(graphite)}$

TC0111

Q.12. Which of the following equations represents a reaction that provides the enthalpy of formation of $CH_3Cl(g)$?

(A) $C(s) + HCl(g) + H_2(g) \rightarrow CH_3Cl(g)$ (B) $C(s) + 3/2\,H_2(g) + 1/2\,Cl_2(g) \rightarrow CH_3Cl(g)$

(C) $C(s) + 3\,H(g) + Cl(g) \rightarrow CH_3Cl(g)$ (D) $CH_4(g) + Cl_2(g) \rightarrow CH_3Cl(g) + HCl(g)$

TC0112

Q.13 The heats of formation of $CO_2(g)$ and $H_2O(l)$ are –394 kJ/mole and –285.8 kJ/mole respectively Using the data for the following combustion reaction, calculate the heat of formation of $C_2H_2(g)$.

$2\,C_2H_2(g) + 5\,O_2(g) \rightarrow 4\,CO_2(g) + 2\,H_2O(l)$; $\Delta H^o = -2601$ kJ

(A) – 238.6 kJ/mole (B) 253.2 kJ/mole (C) 238.7 kJ/mole (D) 226.7 kJ/mole

TC0113

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\e\4-TD-01-TC-TD-02 & 15_{PC}\Eng\2-Thermochemistry\02_Exp6.5

Q.14 The heats of formation of $CO_2(g)$ and $H_2O(l)$ are –394 kJ/mole and –285.8 kJ/mole respectively & $\Delta H^\circ_{Combustion}$ $[C_3H_8(g)] = -2221.6$ kJ. Then the heat of formation of $C_3H_8(g)$ is -

(A) 212.2 kJ/mole (B) – 143.3 kJ/mole (C) 185.4 kJ/mole (D) – 103.6 kJ/mole

TC0114

Enthalpy of formation and combustion

Q15. The enthalpies of combustion of carbon and carbon monoxide are –393.5 and –283 kJ mol^{-1} respectively. The enthalpy of formation of carbon monoxide per mole :- **[AIEEE-2004]**

(A) 110.5 kJ (B) 676.5 kJ (C) –676.5 kJ (D) –110.5 kJ

TC0115

Q.16 Study the following thermochemical equations :

$A \rightarrow B$; $\Delta H = +100$ kcal

$B \rightarrow C$; $\Delta H = -80$ kcal

The correct order of enthalpies of formation of A, B and C is -

(A) A < B < C (B) A < C < B (C) C < A < B (D) B < C < A

TC0036

Q.17 $N_2(g) + 2O_2(g) \rightarrow 2NO_2$ $\Delta H = +x$ kJ

$2NO(g) + O_2(g) \rightarrow 2NO_2(g)$ $\Delta H = +y$ kJ

The enthalpy of formation of NO is

(A) (2x – 2y) kJ/mol (B) (x – y) kJ/mol (C) $\frac{1}{2}(y-x)$ kJ/mol (D) $\frac{1}{2}(x-y)$ kJ/mol

TC0037

Bond enthalpy

Q.18 The enthalpy change for the following reaction is 513 kJ. Calculate the average Cl – F bond energy.

$ClF_3(g) \rightarrow Cl(g) + 3F(g)$

(A) 1542 kJ/mole (B) 88 kJ/mole (C) 171 kJ/mole (D) 514 kJ/mole

TC0116

Q.19 If enthalpy change for hydrogenation of ethylene is –132 kJ/mole and enthalpy of formation 1,3-butadiene (g) and butane (g) are 115 kJ and –140 kJ/mole respectively then calculate resonance energy of 1,3-budadiene (in kJ).

(A) 9 (B) 18 (C) 4 (D) 10 **TC0046**

Q.20 The bond dissociation energy of gaseous H_2, Cl_2 and HCl are 104, 58 and 103 kcal mol^{-1} respectively. The enthalpy of formation for HCl gas will be :-

(A) –44.0 kcal/mol (B) –22.0 kcal/mol (C) 22.0 kcal/mol (D) 44.0 kcal/mol

TC0042

Q.21 The reaction $CH_4(g) + Cl_2(g) \longrightarrow CH_3Cl(g) + HCl(g)$ has $\Delta H = -25$ kcal.

Bond	Bond Enthalpy kCal
$\varepsilon_{C—Cl}$	84
$\varepsilon_{H—Cl}$	103
$\varepsilon_{C—H}$	x
$\varepsilon_{Cl—Cl}$	y
x : y = 9 : 5	

From the given data, what is the bond enthalpy of Cl—Cl bond

(A) 70 kcal/mol (B) 80 kcal/mol (C) 67.75 kcal/mol (D) 57.86 kcal/mol

TC0043

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Q.22. The standard enthalphy of formation of NH_3 is –46.0 kJ mol^{-1}. If the enthalpy of formation of H_2 from its atoms is –436 kJ mol^{-1} and that of N_2 is –712kJ mol^{-1}, the average bond enthalpy of N–H bond in NH_3 is :-

(A) –1102 kJ mol^{-1} (B) –964 kJ mol^{-1} (C) + 352 kJ mol^{-1} (D) +1056 kJ mol^{-1}

TC0117

Q.23 If x_1, x_2 and x_3 are enthalpies of H–H, O=O and O–H bonds respectively, and x_4 is the enthalpy of vaporisation of water, estimate the standard enthalpy of combustion of hydrogen

(A) $x_1+\frac{x_2}{2}-2x_3+x_4$ (B) $x_1+\frac{x_2}{2}-2x_3-x_4$ (C) $x_1+\frac{x_2}{2}-x_3+x_4$ (D) $2x_3-x_1-\frac{x_2}{2}-x_4$

TC0044

Q.24. If the bond dissociation energies of XY, X_2 and Y_2 (all gaseous diatomic molecules) are in the ratio of 1 : 1 : 0.5 and $\Delta_f H$ for the formation of XY is –200 kJ mol^{-1}. The bond dissociation energy of X_2 will be

(A) 200 kJ mol^{-1} (B) 100 kJ mol^{-1} (C) 800 kJ mol^{-1} (D) 300 kJ mol^{-1}

TC0118

Q.25 If bond enthalpy of C–C and C = C are 348 kJ /mole and 615 kJ/mole respectively then calculate enthalpy change (in kJ/mole) which occurs during the isomerisation of cyclopropane (g) into propene(g)

(A) –267 (B) 81 (C) –81 (D) 267 **TC0045**

Q.26 The enthalpy changes for the following processes are listed below :

$Cl_2(g) = 2Cl(g)$, 242.3 kJ mol^{-1}

$I_2(g) = 2I(g)$, 151.0 kJ mol^{-1}

$ICl(g) = I(g) + Cl(g)$, 211.3 kJ mol^{-1}

$I_2(s) = I_2(g)$, 62.76 kJ mol^{-1}

Given that the standard states for iodine and chlorine are $I_2(s)$ and $Cl_2(g)$, the standard enthalpy of formation for ICl(g) is :-

(A) –16.8 kJ mol^{-1} (B) +16.8 kJ mol^{-1} (C) +244.8 kJ mol^{-1} (D) –14.6 kJ mol^{-1}

TC0119

Q.27 ΔH_f^0 of water is – 285.8 kJ mol^{-1}. If enthalpy of neutralisation of monoacid strong base and mono basic strong acid is –57.3 kJ mol^{-1}, ΔH_f^0 of OH^- ion will be

(A) – 228.5 kJ mol^{-1} (B) 228.5 kJ mol^{-1} (C) 114.25 kJ mol^{-1} (D) –114.25 kJ mol^{-1}

TC0049

Q.28 Oxidising power of chlorine in aqueous solution can be determined by the parameters indicated below:

$\frac{1}{2}Cl_2(g) \longrightarrow Cl(g) \longrightarrow Cl^-(g) \longrightarrow Cl^-(aq)$

The energy involved in the conversion of $\frac{1}{2}Cl_2(g)$ to $Cl^-(aq)$

(Using the data $\Delta_{diss}\ H^{\ominus}_{Cl_2} = 240$ kJ mol^{-1}, $\Delta_{eg}\ H^{\ominus}_{Cl} = -349$ kJ mol^{-1}, $\Delta_{hyd} H^{\ominus}_{Cl^-} = -381$ kJ mol^{-1}) will be :-

(A) –610 kJ mol^{-1} (B) –850 kJ mol^{-1} (C) +120 kJ mol^{-1} (D) +152 kJ mol^{-1}

TC0120

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Q.29 The lattice enthalpy of solid NaCl is 772 $kJmol^{-1}$ and enthalpy of solution is 2 $kJmol^{-1}$. If the hydration enthalpy of Na^+ & Cl^- ions are in the ratio of 3:2.5, what is the enthalpy of hydration of chloride ion?

(A) $-140\ kJmol^{-1}$ (B) $-350\ kJmol^{-1}$ (C) $-351.81\ kJmol^{-1}$ (D) –420 kJ/mol

TC0121

Q.30 The enthalpy changes of the following reactions at 27°C are

$Na(s) + \frac{1}{2}Cl_2(g) \longrightarrow NaCl(s)$ $\Delta_rH = -411$ kJ/mol

$H_2(g) + S(s) + 2O_2(g) \longrightarrow H_2SO_4(l)$ $\Delta_rH = -811$ kJ/mol

$2Na(s) + S(s) + 2O_2(g) \longrightarrow Na_2SO_4(s)$ $\Delta_rH = -1382$ kJ/mol

$\frac{1}{2}H_2(g) + \frac{1}{2}Cl_2(g) \longrightarrow HCl(g)$ $\Delta_rH = -92$ kJ/mol;

from these data, the heat change of reaction at constant volume (in kJ/mol) at 27°C for the process $2NaCl(s) + H_2SO_4(l) \longrightarrow Na_2SO_4(s) + 2HCl(g)$ is (R = 8.3 J/K-mol)

(A) 67 (B) 62.02 (C) 71.98 (D) 64.51 **TC0051**

Q.31 Reactions involving gold have been of particular interest to a chemist . Consider the following reactions,

$Au(OH)_3 + 4\ HCl \longrightarrow HAuCl_4 + 3\ H_2O$, $\Delta H = -28$ kcal

$Au(OH)_3 + 4\ HBr \longrightarrow HAuBr_4 + 3\ H_2O$, $\Delta H = -36.8$ kcal

In an experiment there was an absorption of 0.44 kcal when one mole of $HAuBr_4$ was mixed with 4 moles of HCl . What is the percentage conversion of $HAuBr_4$ into $HAuCl_4$?

(A) 0.5 % (B) 0.6 % (C) 5 % (D) 50 % **TC0052**

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EXERCISE # O-II

Single Correct :

Q.1 What is the ratio of the enthalpy yield on combustion of hydrogen atoms to steam to the yield on combustion of an equal mass of hydrogen molecules to steam?

Given : $H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(g)$; $\Delta H = -242$ kJ

B.E. (H – H) = 436 kJ

(A) 0.80 : 1 (B) 1 : 0.80 (C) 1.80 : 1 (D) 2.80 : 1 **TC0053**

Q.2 For the allotropic change represented by the equation C (graphite) $\longrightarrow$ C (diamond), $\Delta H = 1.9$ kJ. If 6 g of diamond and 6 g of graphite are separately burnt to yield CO_2, the heat liberated in first case is

(A) less than in the second case by 1.9 kJ (B) more than in the second case by 11.4 kJ

(C) more than in the second case by 0.95 kJ (D) less than in the second case by 11.4 kJ

TC0054

Q.3 (i) Cis–2 – butene $\rightarrow$ trans – 2 – butene, ΔH_1

(ii) Cis – 2– butene $\rightarrow$ 1 – butene, ΔH_2

(iii) Enthalpy of combustion of 1–butene, $\Delta H = -649.8$ kcal/mol

(iv) $9\Delta H_1 + 5\Delta H_2 = 0$

(v) Enthalpy of combustion of trans 2 – butene, $\Delta H = -647.0$ kcal/mol.

The value of ΔH_1 & ΔH_2 in Kcal/mole are

(A) –1.0 , 1.8 (B) 1.8, –1.0 (C) –5, 9 (D) –2, 3.6 **TC0055**

Q.4 Hydrazine, a component of rocket fuel, undergoes combustion to yield N_2 and H_2O.

$N_2H_4(l) + O_2(g) \longrightarrow N_2(g) + 2H_2O(l)$

What is the enthalpy change of combustion of N_2H_4 (kJ/mole)

Given Reaction	**ΔH/kJ**
$2NH_3(g) + 3N_2O(g) \longrightarrow 4N_2(g) + 3H_2O(l)$	– 1011 kJ
$N_2O(g) + 3H_2(g) \longrightarrow N_2H_4(l) + H_2O(l)$	– 317 kJ
$4NH_3(g) + O_2(g) \longrightarrow 2N_2H_4(l) + 2H_2O(l)$	– 286 kJ
$H_2(g) + \frac{1}{2}O_2(g) \longrightarrow H_2O(l)$	– 285 kJ

(A) – 620.5 (B) – 622.75 (C) 1167.5 (D) + 622.75 **TC0056**

Q.5 The enthalpy change for the reaction,

$CH_3CHO(g) \rightarrow CH_4(g) + CO(g)$ at 300 K is – 17.0 kJ/mol

Calculate the temperature at which $\Delta_r H$ for the reaction will be zero.

[Given : $C_{p,m}(CH_4,g) = 38$ J/K mol ; $C_{p,m}(CO, g) = 31$ J/K mol & $C_{p,m}(CH_3CHO,g) = 52$ J/K mol]

(A) 1300°C (B) 1027°C

(C) 700°C (D) 754ºC **TC0057**

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More than one may correct

Q.6 Select the correct option -

(A) $\Delta H_f(H(g))$ is equal to $\Delta H_{atomisation}$ of $H_2(g)$

(B) $\Delta H_{BE}(H–H)$ is equal to ΔH_f of $H(g)$

(C) $\Delta H_{BE}(H–H)$ is equal to $\Delta H_{atomisation}$ of $H_2(g)$

(D) $\Delta H_{combustion}$ $[H_2(g)]$ is equal to ΔH_f $[H_2O(l)]$ at 300K **TC0058**

Q.7 Which of the following statement is (are) correct ?

(A) for any reaction $\Delta_r H^\circ = \sum \Delta_f H^0_{product} - \sum \Delta_f H^0_{reactant}$

(B) ΔH_f° of $CO_2(g)$ is same as the $\Delta H^\circ_{comb.}$ of carbon graphite

(C) All exothermic gaseous reactions, $\sum (B.E.)_{reactants} > \sum (B.E.)_{products}$

(D) for a reaction $N_{2(g)} + O_{2(g)} \longrightarrow 2NO_{(g)}$, the heat at constant pressure and the heat at constant volume at a given temperature are same **TC0059**

Q.8 Which of the following do(es) not represent ΔH^0 formation of the product.

(A) $H_2(g) + I_2(s) \longrightarrow 2HI(g)$ (B) $\frac{2}{3} O_3(g) \longrightarrow O_2(g)$

(C) $NH_4^+(g) + Cl^-(g) \longrightarrow NH_4Cl(s)$ (D) $P_4(\text{black}) + 5O_2(g) \longrightarrow P_4O_{10}(s)$

(E) Reaction representing $\Delta H_{combustion}$ of C (graphite). **TC0060**

Q.9 From the following data at 25°C

Reaction	**$\Delta_r H^\circ$ kJ/mol**
$\frac{1}{2} H_2(g) + \frac{1}{2} O_2(g) \longrightarrow OH(g)$	42
$H_2(g) + \frac{1}{2} O_2(g) \longrightarrow H_2O(g)$	–242
$H_2(g) \longrightarrow 2H(g)$	436
$O_2(g) \longrightarrow 2O(g)$	495

Which of the following statement(s) is/are correct:

(A) $\Delta_r H^\circ$ for the reaction $H_2O(g) \longrightarrow 2H(g) + O(g)$ is 925.5 kJ/mol

(B) $\Delta_r H^\circ$ for the reaction $OH(g) \longrightarrow H(g) + O(g)$ is 502 kJ/mol

(C) Enthalpy of formation of H(g) is –218 kJ/mol

(D) Enthalpy of formation of OH(g) is 42 kJ/mol **TC0061**

Match the column :

Q.10

	Column-I		**Column-II**
(A)	$C(\text{graphite}) + O_2(g) \rightarrow CO_2(g)$	(P)	$\Delta H^0_{formation}$
(B)	$C(\text{graphite}) \rightarrow C(\text{gas})$	(Q)	$\Delta H^0_{combustion}$
(C)	$CH_3COOH(aq) + OH^-(aq) \rightarrow CH_3COO^-(aq) + H_2O(\ell)$	(R)	$\Delta H^0_{atomization}$
(D)	$CH_4(g) \rightarrow C(g) + 4H(g)$	(S)	$\Delta H^0_{neutralization}$

TC0062

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MATCH LIST :

Q.11 Match the enthalpy change (ΔH) mentioned in list-II for 16 gm O_2 with the various reaction in list-I.

	List-I		**List-II (ΔH in kJ)**
(P)	$2C_2H_2 + 5O_2(g) \longrightarrow 4CO_2(g) + 2H_2O(l)$, ΔH– 2601 kJ	(A)	–285.8
(Q)	$H_2(g) + 1/2O_2(g) \longrightarrow H_2O(g)$, ΔH= – 285.8 kJ	(B)	–196.75
(R)	$3FeO(s) + 1/2O_2(g) \longrightarrow Fe_3O_4(s)$, ΔH = –302.4 kJ	(C)	–260.1
(S)	$C_{graphite} + O_2(g) \longrightarrow CO_2(g)$, ΔH = –393.5 kJ	(D)	–302.4 **TC0063**

Code :

	P	**Q**	**R**	**S**
(A)	4	1	3	2
(B)	3	1	4	2
(C)	3	4	1	2
(D)	2	3	1	4

Paragraph for Q.12 & Q.13

Bond dissociation enthalpy of the first H–S bond in hydrogen sulphide is 376 kJ/mole. The enthalpies of formation of $H_2S(g)$ and S(g) are –20.0 and 277.0 kJ/mole respectively. The enthalpy of formation of gaseous hydrogen atom is 218 kJ/mole. Using above information, answer following questions :

Q.12 The enthalpy of formation of free radical HS is

(A) 138 kJ/mole (B) –138 kJ/mole (C) –10 kJ/mole (D) 357 kJ/mole

TC0064

Q.13 The bond dissociation enthalpy of the free radical HS is

(A) 138 kJ/mole (B) 276 kJ/mole (C) 357 kJ/mole (D) 376 kJ/mole

TC0064

Paragraph for Q.14 & Q.15

Amount of heat evolved during complete combustion of liquid benzene can be calculated from the following data.

(i) 18 gm of graphite on complete combustion evolve 591 kJ heat

(ii) $\Delta H_f^\circ(H_2O, \ell)$ = – 286 kJ/mol

(iii) The heat of formation of liquid benzene is 50 kJ /mole

Q.14 Heat of formation of $CO_2(g)$ from following data is-

(A) –286 kJ/mole (B) –590 kJ/mole (C) –394 kJ/mole (D) –3268 kJ/mole

TC0065

Q.15 Find heat evolved from combustion of 78 gm benzene

(A) 3272 kJ (B) 6345 kJ (C) 4536 kJ (D) 5364 kJ **TC0065**

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Table Type Question :

	Column-I (Reactions)		**Column-II (Characteristics)**		**Column-III (Characteristics)**
(I)	$C(s) + O_2(g) \rightarrow CO_2(g)$	(i)	ΔH = +ve	(P)	$\Delta H > \Delta E$
(II)	$SO_3(g) \rightarrow SO_2(g) + \frac{1}{2}O_2(g)$	(ii)	Δn_g = +ve	(Q)	$\lvert \Delta H \rvert > \lvert \Delta E \rvert$
(III)	$2CO(g) + O_2(g) \rightarrow 2CO_2(g)$	(iii)	Δn_g = –ve	(R)	$\Delta H < \Delta E$
(IV)	$C(s) + \frac{1}{2}O_2(g) \rightarrow CO(g)$	(iv)	ΔH = –ve	(S)	$\lvert \Delta H \rvert < \lvert \Delta E \rvert$

Q.16 Which of the following is only correct match ?
(A) I , i , P (B) II , ii , Q (C) III , iii , S (D) IV , iv , R
TC0066

Q.17 Which of the following is only incorrect match ?
(A) II , i , P (B) III , iv , R (C) IV , iii , S (D) II , ii , Q
TC0066

Q.18 Which of the following is only correct match ?
(A) I , ii , Q (B) II , iii , Q (C) III , iv , S (D) IV , ii , P
TC0066

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EXERCISE # J-MAIN

1. Consider the reaction : **[AIEEE-2011]**

$4NO_2(g) + O_2(g) \rightarrow 2N_2O_5(g)$, $\Delta_rH = -111kJ$.

If $N_2O_5(s)$ is formed instead of $N_2O_5(g)$ in the above reaction, the Δ_rH value will be :-

(given, ΔH of sublimation for N_2O_5 is 54 kJ mol^{-1})

(1) –165 kJ (2) +54 kJ (3) +219 kJ (4) –219 kJ **TC0067**

2. The value of enthalpy change (ΔH) for the reaction

$C_2H_5OH_{(\ell)} + 3O_{2(g)} \rightarrow 2CO_{2(g)} + 3H_2O_{(\ell)}$

at 27°C is –1366.5 kJ mol^{-1}. The value of internal energy change for the above reaction at this temperature will be :- **[AIEEE-2011]**

(1) –1371.5 kJ (2) –1369.0 kJ (3) –1364.0 kJ (4) –1361.5 kJ **TC0068**

3. The enthalpy of neutralisation of NH_4OH with HCl is –51.46 kJ mol^{-1} and the enthalpy of neutralisation of NaOH with HCl is –55.90 kJ mol^{-1}. The enthalpy of ionisation of NH_4OH is:

[JEE-MAINS (online) 2012]

(1) +107.36 kJ mol^{-1} (2) –4.44 kJ mol^{-1} (3) –107.36 kJ mol^{-1} (4) +4.44 kJ mol^{-1}

TC0069

4. Given **[JEE-MAINS (online) 2013]**

Reaction	**Energy Change (in kJ)**
$Li(s) \longrightarrow Li(g)$	161
$Li(g) \longrightarrow Li^+(g)$	520
$\frac{1}{2}F_2(g) \longrightarrow F(g)$	77
$F(g) + e^- \longrightarrow F^-(g)$	(Electron gain enthalpy)
$Li^+(g) + F^-(g) \longrightarrow LiF(s)$	–1047
$Li(s) + \frac{1}{2}F_2(g) \longrightarrow Li\,F(s)$	–617

Based on data provided, the value of electron gain enthalpy of fluorine would be :

(1) –300 kJ mol^{-1} (2) –328 kJ mol^{-1} (3) –350 kJ mol^{-1} (4) –228 kJ mol^{-1} **TC0070**

5. Given : **[JEE-MAINS (online) 2013]**

(1) $H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(l)$; $\Delta H^\circ_{298K} = -285.9$ kJ mol^{-1}

(2) $H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(g)$; $\Delta H^\circ_{298K} = -241.8$ kJ mol^{-1}

The molar enthalpy of vapourisation of water will be :-

(1) 241. 8 kJ mol^{-1} (2) 527.7 kJ mol^{-1} (3) 44.1 kJ mol^{-1} (4) 22.0 kJ mol^{-1} **TC0071**

6. The standard enthalpy of formation ($\Delta_fH^\circ_{298}$) for methane, CH_4 is– 74.9 kJ mol^{-1}. In order to calculate the average energy given out in the formation of a C–H bond from this it is necessary to know which one of the following? **[JEE-MAINS(online) 2014]**

(1) the dissociation energy of the hydrogen molecule, H_2.

(2) the dissociation energy of H_2 and enthalpy of sublimation of carbon (graphite).

(3) the first four ionisation energies of carbon and electron affinity of hydrogen.

(4) the first four ionisation energies of carbon. **TC0072**

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7. For complete combustion of ethanol, $C_2H_5OH(\ell) + 3O_2(g) \rightarrow 2CO_2(g) + 3H_2O(\ell)$, the amount of heat produced as measured in bomb calorimeter, is 1364.47 kJ mol^{-1} at 25°C. Assuming ideality the Enthalpy of combustion, Δ_cH, for the reaction will be :- **[JEE-MAINS(offline)2014]**

($R = 8.314$ J mol^{-1})

(1) –1460.50 kJ mol^{-1}

(2) – 1350.50 kJ mol^{-1}

(3) – 1366.95 kJ mol^{-1}

(4) – 1361.95 kJ mol^{-1} **TC0073**

8. The heats of combustion of carbon and carbon monoxide are – 395.5 and – 285.5 kJ mol^{-1}, respectively. The heat of formation (in kJ) of carbon monoxide per mole is :-

[JEE-MAINS(offline)2016]

(1) – 110.5 (2) 110.5 (3) 676.5 (4) – 676.5 **TC0074**

9. The enthalpy change on freezing of 1 mol of water at 5°C to ice at –5°C is :

(Given $\Delta_{fus}H = 6$ kJ mol^{-1} at 0°C, $C_P(H_2O, l) = 75.3$ J mol^{-1} K^{-1}, $C_P(H_2O, s) = 36.8$ J mol^{-1} K^{-1})

[JEE-MAINS(online)2017]

(1) 6.56 kJ mol^{-1} (2) 5.81 kJ mol^{-1} (3) 6.00 kJ mol^{-1} (4) 5.44 kJ mol^{-1} **TC0075**

10. Given **[JEE-MAINS(offline)2017]**

$C_{(grahite)} + O_2(g) \rightarrow CO_2(g)$; $\Delta_rH° = -393.5$ kJ mol^{-1}

$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(l)$; $\Delta_rH° = -285.8$ kJ mol^{-1}

$CO_2(g) + 2H_2O(l) \rightarrow CH_4(g) + 2O_2(g)$; $\Delta_rH° = +890.3$ kJ mol^{-1}

Based on the above thermochemical equations, the value of $\Delta_rH°$ at 298 K for the reaction

$C_{(grahite)} + 2H_2(g) \rightarrow CH_4(g)$ will be :-

(1) +74.8 kJ mol^{-1}

(2) +144.0 kJ mol^{-1}

(3) –74.8 kJ mol^{-1}

(4) –144.0 kJ mol^{-1} **TC0076**

11. The combustion of benzene (l) gives $CO_2(g)$ and $H_2O(l)$. Given that heat of combustion of benzene at constant volume is –3263.9 kJ mol^{-1} at 25° C ; heat of combustion (in kJ mol^{-1}) of benzene at constant pressure will be - **[JEE-MAINS(offline)2018]**

($R = 8.314$ JK^{-1} mol^{-1})

(1) –452.46 (2) 3260 (3) –3267.6 (4) 4152.6 **TC0077**

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12. Given **[JEE-MAINS(online)2018]**

(i) $2Fe_2O_3(s) \rightarrow 4Fe(s) + 3O_2(g)$; $\Delta_rG^\circ = +1487.0$ kJ mol^{-1}

(ii) $2CO(g) + O_2(g) \rightarrow 2CO_2(g)$; $\Delta_rG^\circ = -514.4$ kJ mol^{-1}

Free energy change, Δ_rG° for the reaction

$2Fe_2O_3(s) + 6CO(g) \rightarrow 4Fe(s) + 6CO_2(g)$

will be :-

(1) –112.4 kJ mol^{-1} (2) –56.2 kJ mol^{-1}

(3) 168.2 kJ mol^{-1} (4) 208.0 kJ mol^{-1} **TC0078**

13. Given : **[JEE-MAINS(online)2019]**

(i) $C(graphite) + O_2(g) \rightarrow CO_2(g)$; $\Delta rH^\circ = x$ kJ mol^{-1}

(ii) $C(graphite) + \frac{1}{2}O_2(g) \rightarrow CO(g)$; $\Delta rH^\circ = y$ kJ mol^{-1}

(iii) $CO(g) + \frac{1}{2}O_2(g) \rightarrow CO_2(g)$; $\Delta rH^\circ = z$ kJ mol^{-1}

Based on the above thermochemical equations, find out which one of the following algebraic relationships is correct ?

(1) z = x + y (2) x = y – z (3) x = y + z (4) y = 2z – x **TC0079**

14. The difference between ΔH and ΔU ($\Delta H - \Delta U$), when the combustion of one mole of heptane (1) is carried out at a temperature T, is equal to: **[JEE-MAINS-(ONLINE)-2019]**

(1) 3RT (2) –3RT (3) –4RT (4) 4RT **TC0080**

15. Enthalpy of sublimation of iodine is 24 cal g^{-1} at 200°C. If specific heat of $I_2(s)$ and $I_2(vap)$ are 0.055 and 0.031 cal g^{-1}K^{-1} respectively, then enthalpy of sublimation of iodine at 250°C in cal g^{-1} is :

[JEE-MAINS(online)2019]

(1) 2.85 (2) 11.4 (3) 5.7 (4) 22.8 **TC0081**

16. The standard heat of formation $\left(\Delta_f H^0_{298}\right)$ of ethane in (kJ/mol), if the heat of combustion of ethane, hydrogen and graphite are –1560, –393.5 and –286 kJ/mol, respectively is _____

[JEE-MAINS(online)2020]

TC0082

17. If enthalpy of atomisation for $Br_{2(1)}$ is x kJ/mol and bond enthalpy for Br_2 is y kJ/mol, the relation between them : **[JEE-MAINS(online)2020]**

(1) is x = y (2) is x < y

(3) does not exist (4) is x > y **TC0083**

18. If the standard molar enthalpy change for combustion of graphite powder is -2.48×10^2 kJ mol^{-1}, the amount of heat generated on combustion of 1 g of graphite powder is _____ kJ. (Nearest integer)

[JEE-MAINS(online)2021]

TC0122

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19. At 298 K, the enthalpy of fusion of a solid (X) is 2.8 kJ mol^{-1} and the enthalpy of vaporisation of the liquid (X) is 98.2 kJ mol^{-1}. The enthalpy of sublimation of the substance (X) in kJ mol^{-1} is _____ .
(in nearest integer) **[JEE-MAINS(online)2021]**

TC0123

20. For water at 100°C and 1 bar,
$\Delta_{vap}H - \Delta_{vap}U$ = _____ × 10^2 J mol^{-1}. **[JEE-MAINS(online)2021]**
(Round off to the Nearest Integer)
[Use : R = 8.31 J mol^{-1} K^{-1}]
[Assume volume of $H_2O(l)$ is much smaller than volume of $H_2O(g)$. Assume $H_2O(g)$ treated as an ideal gas]

TC0124

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EXERCISE # J-ADVANCE

Q.1 Using the data provided, calculate the multiple bond energy (kJ mol^{-1}) of a C ≡ C bond in C_2H_2. That energy is (take the bond energy of a C–H bond as 350 kJ mol^{-1}.) **[JEE 2012]**

$2\ C(s) + H_2(g) \longrightarrow C_2H_2(g)$ $\Delta H = 225\ kJ\ mol^{-1}$

$2\ C(s) \longrightarrow 2C(g)$ $\Delta H = 1410\ kJ\ mol^{-1}$

$H_2(g) \longrightarrow 2H(g)$ $\Delta H = 330\ kJ\ mol^{-1}$

(A) 1165 (B) 837 (C) 865 (D) 815 **TC0087**

Q.2 The standard enthalpies of formation of $CO_2(g)$, $H_2O(l)$ and glucose(s) at 25°C are –400 kJ/mol, –300 kJ/mol & –1300 kJ/mol, respectively. The standard enthalpy of combustion per gram of glucose at 25°C is - **[JEE 2013]**

(A) + 2900 kJ (B) – 2900 kJ (C) –16.11 kJ (D) + 16.11 kJ **TC0088**

Paragraph For Questions 3 and 4

When 100 mL of 1.0 M HCl was mixed with 100 mL of 1.0 M NaOH in an insulated beaker at constant pressure, a temperature increase of 5.7 °C was measured for the beaker and its contents. **(Expt-1).** Because the enthalpy of neutralisation of a strong acid with a strong base is a constant (–57.0 kJmol^{-1}), this experiment could be used to measure the calorimeter constant. In a second experiment **(Expt-2)**, 100 mL of 2.0 M acetic acid ($K_a = 2.0 \times 10^{-5}$) was mixed with 100 mL of 1.0M NaOH (under identical conditions to **(Expt-1))** where a temperature rise of 5.6 °C was measured.

(Consider heat capacity of all solutions as 4.2 $Jg^{-1}K^{-1}$ and density of all solutions as 1.0 g mL^{-1}) **[JEE 2015]**

Q.3 Enthalpy of dissociation (in kJ mol^{-1}) of acetic acid obtained from the **Expt-2** is

(A) 1.0 (B) 10.0 (C) 24.5 (D) 51.4 **TC0089**

Q.4 The pH of the solution after **Expt-2**

(A) 2.8 (B) 4.7 (C) 5.0 (D) 7.0 **TC0089**

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Q.5 Choose the reaction(s) from the following options, for which the standard enthalpy of reaction is equal to the standard enthalpy of formation. **[JEE 2019]**

(A) $\frac{3}{2}O_2(g) \rightarrow O_3(g)$

(B) $\frac{1}{8}S_8(s) + O_2(g) \rightarrow SO_2(g)$

(C) $2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$

(D) $2C(g) + 3H_2(g) \rightarrow C_2H_6(g)$ **TC0090**

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ANSWER KEY

EXERCISE # S-I

Q.1 178.00 kJ

Q.2 1000.00 cal

Q.3 –741.50 kJ/mole

Q.4 –3.50 kJ

Q.5 – 139.60

Q.6 189.40 kcal

Q.7 –0.30 kJ

Q.8 –80.00

Q.9 –1560.00

Q.10 129.00 kJ

Q.11 –3201, – 3199.75 kJ/mole

Q.12 - 88.00 kJ mol^{-1}

Q.13 –266.00 kJ/mol & –825 kJ/mole

Q.14 –41.20 kJ

Q.15 (i) –885 kJ/mol (ii) – 890 kJ/mole

Q.16 (602.49 kJ/mole)

Q.17 4.50 %

Q.18 400.00 kJ/mole

Q.19 20.60 kcal/mole

Q.20 – 266 kJ/mole

Q.21 –72 kJ/mole

Q.22 201.00 kJ

Q.23 5.00 kJ/mole

Q.24 –2669.00 kJ/mole

Q.25 –84.00 kJ/mole

Q.26 –352.00 kJ/mole

Q.27 –19.00 kJ/mole

Q.28 –80.00 kJ/mole

Q.29 6.00 kJ/mole

Q.30 5.00 kJ/mole

Q.31 31.50 kJ/mole

EXERCISE # S-II

Q.1 22 kcal mol^{-1}

Q.2 (a) 20 g (b) 20.28 g

Q.3 0.93 or 0.94 kcal g^{-1}, 3.94 kcal cm^{-3}

Q.4 292 kcal/mol

Q.5 E(O–H) = 463.5 kJ/mol ΔU = 461 kJ/mol

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Q.6 **(i) 342 kJ/mol^{-1} ; (ii) 892 kJ/mol^{-1}**

Q.7 **2 kcal**

Q.8 **25 kJ/mol**

Q.9 **– 669.7 kJ mol^{-1}**

Q.10 **–123 kJ/mol**

EXERCISE # O-I

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	D	A	B	D	B	D	B	A	D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	B	B	D	D	D	B	D	C	A	B
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	D	C	B	C	B	B	A	A	B	B
Que.	31									
Ans.	C									

EXERCISE # O-II

Que.	1	2	3	4	5	6	7	8	9	
Ans.	D	C	A	A	B	C,D	A,B,D	A,B,C,D	A,D	
Que.	10				11	12	13	14	15	16
Ans.	A-P,Q	B-P,R	C-S	D-R	B	A	C	C	A	B
Que.	17	18								
Ans.	C	D								

EXERCISE # J-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	4	3	4	2	3	2	3	1	1	3
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	3	2	3	3	4	-192.50	4	21	101	31

EXERCISE # J-ADVANCED

Que.	1	2	3	4	5	
Ans.	D	C	A	B	A,B	

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